

Danielle Grau Ph.D.

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🔗 <https://dgrau13.github.io/>

🔗 <https://github.com/dgrau13>

EDUCATION

- **Georgia Institute of Technology**
Doctor of Philosophy - Earth & Atmospheric Sciences; GPA: 3.87 Aug 2021 - June 2026
Doctoral Minor: High Performance Computing
- **Florida International University**
Bachelors of Science - Physics; GPA: 3.92 Jan 2018 - May 2021
Honors: Summa Cum Laude, ΦBK

EXPERIENCE

- **Graduate Research Assistant**
Georgia Institute of Technology Aug 2021 - July 2026
 - **Analyze Observational, Remote-Sensing Datasets:** Developed MATLAB and Python-based workflows to analyze the statistical surface roughness properties of the entire Antarctic and Greenland Ice Sheets from NASA ICESat-2 Satellite Altimetry tracks.
 - **Designed and Parallelized Monte-Carlo Simulations:** Developed a MATLAB-based workflow that generates a random, statistically rough surface and uses a cellular automaton model to simulate surface water distribution and routing.
 - **Extrapolate Relationships from Large Simulated Datasets:** Utilized MATLAB's Curve Analysis library to derive mathematical relationships from a Monte-Carlo simulated data set.
 - **Model Configuring and Tuning:** Performed Transient Projections for Antarctic Ice Sheet Loss from 2015 to 2300 with external GCM and Ocean Forcing.
 - **Large Ice-Sheet Model Component Development:** Developed new class and calving scheme within NASA's Ice-Sheet and Sea-Level System Model (ISSM) using MATLAB and C++.
 - **Present Scientific Research:** Present academic research in both poster and oral format at national and international conferences.
 - **Scientific Writing:** Composed and published scientific articles in high-profile journals (*Nature Communications*)
 - **Collaboration:** Collaborated with other professional scientific researchers from other research institutions.
 - **Mentor Undergraduate Researchers:** Provide resources and guidance on research projects for three undergraduate student researchers.
 - **Secured over \$50,000 in fellowships and grants:** Internal fellowships & the NASA Georgia Space Grant Consortium.
- **President, Graduate Students in Earth & Atmospheric Sciences**
Georgia Institute of Technology, School of Earth & Atmospheric Sciences June 2024 - May 2025
 - **Delegation:** Collaborate with other members of the executive board and assign tasks.
 - **Organize Events:** Graduate Student Socials, Professional Development Events, Department-wide events.
 - **Coordinate with Department Faculty and Staff:** Deliver concerns and worries from graduate students and collaborate on solutions.
- **Undergraduate Research Assistant**
Georgia Institute of Technology May 2020 - May 2021
 - **Signal and Power Spectral Density Analysis:** Developed a Python-based workflow to perform PSD analysis on ICESat-2 land-ice Altimetry tracks.
- **Undergraduate Teaching Assistant**
Florida International University Jan 2020 - May 2021
 - **Lecture Aid:** Aid students' comprehension of course materials and host exam review sessions.
 - **Laboratory Aid:** Assist students in laboratory experiments and analysis for Intro. Physics I & II Lab sections.
 - **Exam Proctor:** Assist in proctoring and prepping exam materials.

SKILLS SUMMARY

- **Languages:** Python, MATLAB, Linux, C++, L^AT_EX, Mathematica
- **Frameworks:** Scikit, Pandas, TensorFlow, Keras, PBS, Slurm
- **Tools:** GitHub, Machine Learning, Parallel Computing, Cloud Computing, High Performance Computing, Data Visualization, MS Office, Adobe Illustrator
- **Soft Skills:** Leadership, Event Organization & Management, Scientific & Professional Writing, Public Speaking
- **Linguistics:** English, Spanish, French

PROJECTS

- **NASA Goddard Institute of Space Studies:** Implementing supraglacial melt lake parameterizations into existing albedo scheme for glacier ice in ModelE. (Feb 2024)
- **Patagonian Ice-field Research Program (PIRP):** Measured Calving Events at Grey Glacier, Chile using an Aquarian Audio & Scientific H2d Hydrophone faster. Data was analyzed using signal processing in the field. Tech: MATLAB, Signal Processing. (Nov 2023)
- **ICESat-2 Hackweek 2023:** Develop a Python-based workflow to analyze grounding zone features in Antarctica using NASA's ICESat-2 Land Ice Altimetry. Tech: CryoCloud, SlideRule Earth (Aug 2023)

COMMUNITY ENGAGEMENT & LEADERSHIP EXPERIENCE

- **Graduate Student Senator** Georgia Tech
Graduate Senator for the School of Earth and Atmospheric Sciences Aug 2025 - Present
 - **Review and Vote on Legislation:** Review and Vote on incoming bills from various committees.
- **President of $\Sigma\Pi\Sigma$ Florida International University**
National Physics and Astronomy Honor Society May 2020 - Apr 2021
 - **Delegation:** Collaborate with other members of the executive board.
 - **Organize & Host Events:** Professional Development Events, Career Advice, Coding Workshops
- **APS Inclusion, Diversity, and Equity Alliance (Florida International University Member)**
American Physical Society Aug 2020 - Apr 2021
- **Vice President of $\Sigma\Pi\Sigma$ Florida International University**
National Physics and Astronomy Honor Society May 2019 - Apr 2020
 - **Coordinate Conference Travel:** Arrange transport, lodging and conference registration for APS March and SEASAPS for physics undergraduate students.
 - **Oversaw Pion Mentorship Program:** Matched senior undergraduate students and junior undergraduate students in mentor-mentees pairings. Organized events for mentors and mentees including professional development, navigating university, and academic advising.
- **Secretary of $\Sigma\Pi\Sigma$ Florida International University**
National Physics and Astronomy Honor Society Sep 2018 - Apr 2019
 - **Record-Keeping:** Kept meeting notes and extensive records of organization meetings and events. Allow kept attendance of all events and meetings.
 - **Communication:** Main point of contact for the organization for both undergraduate students and university departments.
 - **Marketing and Promotion:** Oversaw market and promotions of organizations events and meetings via social media.

PUBLICATIONS

1. **Grau, D.**, M. Poinelli, N.J. Schlegel, H. Seroussi, A.A. Robel. Accelerated Ice Loss from Antarctica in Simulations with Calving Driven by Supraglacial Melt Lake Depth. (Submitted *Geophysical Research Letters*, June 2026).
2. **Grau, D.**, A. Hussain, A.A. Robel (2025). Predicting Mean Depth and Area Fraction of Antarctic Supraglacial Melt Lakes with Physics-Based Parameterizations, *Nature Communications*, In Press.